

Practice NumPy:-

1. Generate **6 random float numbers** between 0 and 1 using `np.random.rand()`.
 2. Create a **2×5 random float matrix** using `np.random.rand()`.
 3. Generate **8 random integers** between **10 and 20** using `np.random.randint()`.
 4. Create a **3×3 matrix** of random integers between **1 and 9**.
 5. Generate **5 random numbers** from **standard normal distribution** using `np.random.randn()`.
-

6. Create a random array of shape **(4, 3, 2)** using `np.random.rand()` and:
 - print its `shape`
 - print its `ndim`
 7. Generate a 1D random integer array of size **15** and find:
 - maximum value
 - minimum value
 - mean value
 8. Create a random 2D integer array (4×4) and check:
 - number of rows
 - number of columns
 - total elements
-

9. Given:

```
10. arr = np.array([10, 20, 30, 40, 50])
```

Pick **3 random elements** using `np.random.choice()`.

11. From the same array, pick **4 elements with repetition**.

12. Shuffle this array using `np.random.shuffle()`:

```
arr = np.array([1, 2, 3, 4, 5, 6])
```

12. Use `np.random.seed(10)` and generate **5 random integers (1–50)**.
Run the code twice and verify the output is the same.

13. **Student Marks**

- Generate marks for **12 students** (0–100)
- Find:
 - highest mark
 - lowest mark
 - average mark
- Randomly select **3 students** for a prize

14. Weekly Sales

- Generate sales for **7 days** (200–1000)
- Find:
 - highest sales day
 - lowest sales day
- Shuffle the sales data

15. Dice Game

- Simulate rolling a dice **20 times**
- Count how many times **6** appears

16. Generate a **5×5** random integer matrix (10–99) and find:

- max value
- min value
- mean of all elements

17. Create random salaries for **8 employees** (30,000–80,000) and:

- find lowest salary
- randomly pick 2 employees for bonus

18. Create an array of **100 random integers** and:

- find how many are greater than 50

19. What is the difference between:

- `np.random.rand()`
- `np.random.randn()`

20. Why do we use `np.random.seed()`?

